

Exam Date & Time: 19-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Computer Applications (BCA)
Semester-I University Examination November 2019

Date: 19/11/2019, Tuesday

Time: 10:00 a.m. to 01:00 p.m.

FOUNDATION OF MATHEMATICS [CA114]

Marks: 70

Duration: 180 mins.

Section-I

Answer all the questions.

Section Duration: 30 mins

Q.1)

If $U = \{1,2,3,4,5\}$ $A = \{1,2,5\}$ $B = \{3,4\}$, then $A \cap B' = \underline{\hspace{2cm}}$

(1)

1)

1) $\{3,4\}$ 2) $\{1,2,5\}$ 3) $\{1,2,3,4,5\}$ 4) $\{\emptyset\}$

2)

$f(x) = 2x^2 - 2x + 4$, $Df = \{1\}$. Find $Rf = \underline{\hspace{2cm}}$.

(1)

1) $\{4\}$ 2) $\{-5\}$ 3) $\{5\}$ 4) $\{-4\}$

3)

What is $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix} = ?$

(1)

1) $\begin{bmatrix} 4 & 3 \\ 4 & 8 \end{bmatrix}$ 2) $\begin{bmatrix} 0 & 3 \\ 4 & 2 \end{bmatrix}$ 3) $\begin{bmatrix} -4 & 3 \\ 4 & -8 \end{bmatrix}$ 4) $\begin{bmatrix} 0 & 3 \\ 4 & -2 \end{bmatrix}$

4)

If $y = \sin x$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

(1)

1) $-\cos x$ 2) $\cos x$

5)

What is the correct factor of $2x^2 + 7x + 3$?

(1)

1) $(3x+1)(x+2)$ 2) $(2x+3)(x+1)$ 3) $(3x+2)(x+1)$ 4) $(2x+1)(x+3)$

6) What is integration of a^x ?

- 1) a^x 2) $a^x \log a$ 3) $\frac{a^x}{\log a}$ 4) None of the given

(1)

7) Which of the following two sets are equal?

- 1) $A = \{1, 2\}$ and $B = \{1\}$ 2) $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ 3) $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$ 4) $A = \{1, 2, 4\}$ and $B = \{1, 2, 3\}$

(1)

8)

If $U = \{1, 2, 3, 4, 5\}$ $A = \{1, 2, 5\}$ $B = \{3, 4\}$ then $A \cap B' = A \cup B'$

- 1) False 2) True

(1)

9) A set that does not contain any element is called ____ set.

- 1) single 2) zero 3) none 4) null

(1)

10)

Identify the type of matrix. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- 1) Diagonal Matrix 2) Identity Matrix 3) Scalar Matrix 4) Square Matrix

(1)

11)

What is transpose of matrix $\begin{bmatrix} -3 & 1 & 4 \\ 5 & 3 & 2 \end{bmatrix}$?

- 1) $A^T = \begin{bmatrix} -3 & 5 \\ 1 & 3 \\ 4 & 2 \end{bmatrix}$ 2) $A^T = \begin{bmatrix} -5 & -3 & -2 \\ 3 & -1 & -4 \end{bmatrix}$ 3) $A^T = \begin{bmatrix} 3 & -1 & -4 \\ -5 & -3 & -2 \end{bmatrix}$ 4) $A^T = \begin{bmatrix} 3 & -5 \\ -1 & -3 \\ -4 & -2 \end{bmatrix}$

(1)

12) What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?

- 1) $\{(1, a), (1, b), (2, a), (2, b)\}$ 2) $\{(1, 1), (2, 2), (a, a), (b, b)\}$ 3) $\{(1, a), (2, a), (1, b), (2, b)\}$ 4) $\{(1, 1), (a, a), (2, a), (1, b)\}$

(1)

13) Integration is the reverse process of differentiation.

- 1) False 2) True

(1)

14)

Which of the following is the derivative of $y = x^5$?

(1)

1)	$5x^3$	2)	$5x^4$	3)	$\frac{x^6}{6}$	4)	$4x^5$
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15)

The process of finding the derivative of a function is known as _____.

(1)

1) limit 2) Derivation 3) Differentiation 4) Integration

16)

If $y = e^x$ then $\frac{dy}{dx} = e^x$.

(1)

1) True 2) False

17)

The value of $\lim_{x \rightarrow 2} 5x + 3 =$ _____.

(1)

1)	13	2)	3	3)	5	4)	2
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18)

 $A \cup A' =$ _____.

(1)

1)	$\emptyset$	2)	U
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19)

Find determinant value of $A = \begin{vmatrix} 1 & 3 & 0 \\ 0 & 2 & 3 \\ 1 & 3 & 2 \end{vmatrix}$

(2)

1) -4 2) 4 3) 14 4) -14

Section-II**Answer all the questions.**

Q.2)

Let $U = \{a, b, c, d, \dots, h\}$, $A = \{a, c, d, e, f\}$, $B = \{a, b, c, f, g\}$, then prove that

$$A \Delta B = (A - B) \cup (B - A).$$
(4)

A)

[OR]

Let $U = \{1, 2, 3, \dots, 10\}$, $A = \{x : x \in N, x \text{ is odd number and } 1 < x \leq 10\}$,

B)

 $B = \{y : y \in N, y \text{ is divisible by 3 and } 1 \leq y < 10\}$, then prove that $(A \cup B)' = (A' \cap B')$

(4)

C)

 $f(x) = 3x^2 + 5$, $g(x) = 2x - 1$, where $x = \{1, 2\}$. Check whether $f(x) = g(x)$.

(4)

[OR]

D)

If $g(x) = \frac{x^2 - 2x}{x - 1}$, then find $g(-1) + g(3) + g(4)$.

(4)

E)

(4)

Discuss the continuity of the function at $x = 2$. $f(x) = \begin{cases} x - 2, & 0 < x < 2 \\ x^3 - 2x, & x = 2 \\ x + 2, & x > 2 \end{cases}$

[OR] Evaluate following equations.

F)

(i) $\lim_{x \rightarrow 4} \frac{\frac{1}{4} + \frac{1}{x}}{4 + x}$ (ii) Find $\lim_{x \rightarrow 2} 5x + 2$ by table method. (4)

Q.3)

Find $\lim_{x \rightarrow 3} \frac{x^3 - 4x^2 - 3x + 18}{x^3 - 8x^2 + 21x - 18}$ (6)

A)

[OR] Evaluate following equations.

B)

(i) $\lim_{n \rightarrow \infty} \frac{5n^2 - 4n + 10}{2n + 5n^2 - 20}$ (ii) Find the value of x , if $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 8x + 12} = \frac{x}{-4}$ (6)

C) Let $U = \{a, b, c, d, \dots, h\}$, $P = \{a, c, d, e\}$, $Q = \{b, c, e, f\}$, $R = \{a, c, e, h\}$.

Find (i) P' (ii) R' (iii) $R' - P$ (iv) $P' \cap Q$ (4)

[OR] Define following term with example:

D)

(i) Finite Set (ii) Singleton Set (iii) Powe Set (iv) complement of set (4)

E) If $f(x) = x^2 + 4x + 5$ and $g(x) = 2x + 1$, prove that $f(1) - 5g(0) - g(2) = 0$. (3)

[OR] Define following term with example:

F)

(i) Relation (ii) Function (iii) Many to one Relation (3)

Section-III

Answer all the questions.

Q.4)

Find inverse of $A = \begin{bmatrix} 5 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 3 & 1 \end{bmatrix}$ (4)

A)

[OR]

B)

Find AB , if $A = \begin{bmatrix} 5 & 3 & 0 \\ 1 & 0 & 3 \\ 0 & 3 & 2 \end{bmatrix}$ & $B = \begin{bmatrix} 4 & 3 & 0 \\ 5 & 0 & 3 \\ 0 & 3 & 2 \end{bmatrix}$ (4)

C) If $f(x) = 4x^3 + 2x^2 + 5x - 2$, then find $f'(3)$ and $f''(2)$. (4)

[OR]
D) Find derivative: $y = \frac{x^2 - x + 1}{x^2 + x - 1}$ (4)

E) Evaluate: $\int_1^4 (5x^2 - 8x + 5) dx$ (4)

[OR]
F) Evaluate following:
(i) $\int (7x^3 + 3x^2 + 2x + 5) dx$ (ii) $\int (2^x - 2e^x + \frac{2}{x} + 15) dx$ (4)

Q.5) Solve the system with three variables by Cramer's Rule.

A)
$$\begin{aligned} -2x + y + z &= 4 \\ -4x + 2y - z &= 8 \\ -6x - 3y + z &= 0 \end{aligned}$$
 (5)

[OR]
B) Find the value of D, if $D = A + 3I - C$ where $C = 2B$, $B = A^T$ and $A = \begin{bmatrix} 0 & 1 & 2 \\ 5 & 0 & -2 \\ 0 & 3 & 1 \end{bmatrix}$ (5)

C) Find the area bounded by $y = \frac{1}{x-2} - \frac{1}{x^3}$, the x-axis on the line $x = 1$ and $x = 2$. (4)

[OR]
D) Find derivative of following functions.
(i) $y = 2^x e^x$ (ii) $y = 2\sqrt{x} - 2e^x + \frac{2}{x} + 15$ (4)

E) If $y = (3x^2 + 5)(2x + 7)$, find $f'(3)$ (4)

[OR]
F) If $y = \frac{(5x+3)}{(3x+2)}$, Find derivative of it and find value when $x = 4$. (4)

-----End-----